MILITARY SPECIFICATION

DETONATOR, ELECTRIC M78, PARTS AND LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing of one type of electric detonator with a powdered iron plug assembly designated as the M78 Electric Detonator.

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-I-559 - Ink, Marking, Stencil, Opaque, for Porous Surfaces, Wood Boxes, Fiber Cartons, etc.

MILITARY

MIL-P-116 - Packaging and Packing for Overseas Shipment
MIL-B-117 - Bags, Interior Packaging
MIL-A-148 - Aluminum Foil
MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment Supply and Maintenance of

FSC: 1345

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-650 - Explosives: Sampling, Inspection and Testing
MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
9206140 - Detonator, Electric, M78 Assembly

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET 9206140 - Equipment Tabulation Number

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other Publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

INTERSTATE COMMERCE COMMISSION

49 CFR 71-90 - Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

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(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-90 (latest revision)".)

3. REQUIREMENTS

3.1 Material.- Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Detonators.- The detonators shall comply with all requirements specified on Drawing (dwg.) 9206140, and with all requirements specified in applicable specifications.

3.3 Moisture content.

3.3.1 Lead azide.- The moisture content of the lead azide at the loading station at the time of loading the detonator shall not exceed 0.50 percent when tested as specified in 4.4.1.1.

3.3.2 Pentaerythrite tetranitrate (PETN).- The moisture content of the PETN at the loading station at the time of loading the detonators shall not exceed 0.50 percent when tested as specified in 4.4.1.2.

3.4 Functioning.- The detonators shall comply with the following:

3.4.1 Time.- The detonators shall fire on the first attempt within 120 microseconds when tested as specified in 4.4.2.1.

3.4.2 Output.- The detonators shall fire on the first attempt and shall make a 3/16 inch minimum (min.) perforation in a .1345 inch min. thick lead disc when tested as specified in 4.4.2.2.

3.4.3 Detonators that fail to fire on the first attempt (Duds).- The detonators shall not fail to fire on the first attempt when tested as specified in 4.4.2.3.

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3.5 Waterproofness.- The detonator assembly shall comply with the requirements of 3.4.2 and 3.4.3 when tested as specified in 4.4.2.2.

3.6 Workmanship.- All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.7 First article inspection.- This specification makes provisions for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-255C shall apply.

4.1.1 Submission of product.- At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspection performed.

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d. Drawing, specification number and date, together with an identification and date of changes.

e. Statement of findings on all explosives and associated ingredients, and certificates of conformance on all remaining material when such material is covered by Government specifications.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.- Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First article inspection.

4.2.1 Submission.- The contractor shall submit a first article sample to an approved laboratory designated by the contracting officer. The first article sample shall consist of twenty-five (25) complete sets of parts which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials shall be obtained from the same source of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance for materials (see 4.1.1.e). A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there

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is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 Inspections to be performed.- Components will be subjected by the Government to any or all of the examinations specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.- If any component fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions.

4.3.1 Lot formation.- The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting

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safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Each lot shall contain:

a. Parts of one lot interfix number from one manufacturer.

b. Lead azide from one batch or blending of batches which results in homogeneity of all lead azide used in loading the entire detonator lot.

c. PETN from one batch or blending of batches which results in homogeneity of all PETN used in loading the entire detonator lot.

Inspection lots shall comply with Standard MIL-STD-105 and shall be numbered in accordance with Standard MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period, nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components, or sub-assemblies other than items for delivery, shall be homogeneous and of a size convenient to the contractor.

4.3.2 Examinations.- Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects (and Majors, when so specified) shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government and shall provide, as a minimum, the protection afforded the Government by the sampling plans in Standard MIL-STD-105. Continuous sampling plans in accordance with Standard MIL-STD-1235 for defects other than critical may be used if approved by the procuring activity. Also, at the option of the procuring activity AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Cup (see dwg. 9206134 covering a detail of dwg. 9206140).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Split, cut or crack through cup....	Visual	01001
Major:	AQL 0.40 percent		
101.	Thickness through bottom maximum (max.).....	Gage	01002
102.	Cup dented or otherwise distorted.	Visual	01003
Minor:	AQL 1.00 percent		
201.	Inside diameter, minimum (min.)..	Gage	01004
202.	Evidence of poor workmanship (see 3.6).....	Visual	01005

4.3.2.2 Ferrule (see dwg. 9206133 covering a detail of dwg. 9206140).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Inside diameter.....	Gage	02001
Minor:	AQL 1.50 percent		
201.	Outside diameter.....	Gage	02002
202.	Length.....	Gage	02003
203.	Evidence of poor workmanship (see 3.6).....	Visual	02004
204.	Chamfer missing.....	Visual	02005

4.3.2.3 Attenuator plug and wire assembly (see dwg. 9206137 covering a detail of dwg. 9206140).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			

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Major:	AQL 1.00 percent		
101.	Distance between wires at bridge end	Gage	03001
102.	Distance between wires at lead end (max.)	Gage	03002
103.	Outside diameter of plug	Gage	03003
104.	Plug cracked or chipped	Visual	03004
105.	Lead wire nicked, missing, or broken	Visual	03005
Minor:	AQL 2.50 percent		
201.	Length of wires at tinned end	Gage	03006
202.	Protrusion of wires, min	Gage	03007
203.	Solder on wire less than min. distance from plug	Gage	03008
204.	Length of plug less boss	Gage	03009
205.	Height of plug boss	Gage	03010
206.	Diameter of plug boss	Gage	03011
207.	Evidence of poor workmanship (see 3.6)	Visual	03012

4.3.2.4 Bridge and wire assembly, prior to application of bridge wire (see dwg. 9206139 covering a detail of dwg. 9206140).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Length of bridge posts	Gage	04001
102.	Molding flash not removed from bridge post	Visual	04002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	04003

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4.3.2.5 Bridge and wire assembly (see dwg. 9206139 covering a detail of dwg. 9206140).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Spot charge missing.....	Visual	05001
102.	Plug cracked or assembly damaged to extent that function may be impaired.....	Visual	05002
103.	Bridge wire not totally covered by spot charge.....	Visual	05003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	05004

4.3.2.6 Assembly, prior to inserting bridge and wire assembly, ferrule and priming charge (see dwg. 9206140).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Height of charge.....	Gage	06001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	06002

4.3.2.7 Assembly (see dwg. 9206140).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Lead wires not shunted.....	Visual-Manual	07001
2.	Cup cracked or split.....	Visual	07002

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Major:	AQL 0.65 percent		
101.	Length	Gage	07003
102.	Diameter	Gage	07004
103.	Crimp not full 360 degrees or not flat against plug	Visual	07005
104.	Coating insufficient	Visual	07006
Minor:	AQL 1.00 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	07007
202.	Marking missing, incorrect or illegible	Visual	07008

4.3.2.8 Carton, prior to sealing (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Incorrect number of detonators	Visual	08001
102.	Detonators in bundle not completely encased in foil	Visual	08002
103.	Packet assembly torn or insecurely closed (when applicable)	Visual	08003
Minor:	AQL 1.00 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	08004
202.	Package marking missing, incorrect or illegible	Visual	08005

4.3.2.9 Carton, sealed (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 percent		
101.	Carton damaged to extent that contents are exposed or liable to become exposed.....	Visual	09001
Minor:	AQL 0.65 percent		
201.	Marking missing, incorrect, or illegible.....	Visual	09002

4.3.2.10 Sealed carton in sealed barrier bag (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Bag punctured or separation of heat seal.....	Visual	10001
Minor:	AQL 0.65 percent		
201.	Marking missing, incorrect or illegible.....	Visual	10002

4.3.3 Production testing.- Unless otherwise specified herein, the sampling plans for the following tests shall be in accordance with Standard MIL-STD-105.

4.3.3.1 Moisture content (see 3.3) - Major defect

Table I

Explosives	Code No.
Lead Azide	11001
PETN	11002

The contractor shall provide adequate controls to insure that the explosives comply with the requirement. For verification, the contractor shall select and subject to test, a sample from that quantity of each explosive used in each eight hours of loading production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirements and loading has not

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begun, the explosive represented by the sample shall be rejected. If components have been loaded with explosive containing excessive moisture, the remaining unloaded explosive and the loaded components shall be rejected. The test procedures shall be as specified in 4.4.1.1 and 4.4.1.2.

4.3.3.2 Functioning and waterproofness.

4.3.3.2.1 Sampling.- One hundred and eighty-five (185) detonator assemblies shall be selected from each lot for test. The sequence of testing and number of samples to be subjected to each test is specified in Table II.

Table II

Sequence	Test	Classification	Code No.	No of Samples	Test Paragraph
1.	Time (see 3.4.1)	Major	07009	35	4.4.2.1
2.	Waterproofness and Output (see 3.4.2, 3.5)	Major	07010	50	4.4.2.2
3.	Duds (see 3.4.3)	Major	07011	*100 less time and output duds	4.4.2.3

*Duds that occur in the time or output phase of testing shall be recorded and replaced from this sample.

4.3.3.2.2 Rejection.- The lot shall be rejected if any of the following test results occur:

a. Time.- The mean functioning time ($\bar{X}$) plus the product of 2.27 and the standard deviation (s) (see 6.5) exceeds the requirement specified in 3.4.1.
b. Waterproofness and Output.- One or more detonator assemblies fail to produce a perforation of the required size in the lead disc.
c. Duds.- The total number of detonator assemblies that fail to fire on the first attempt (dud) in the time, output and dud tests, equals or exceeds two (2).

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4.3.3.3 Pull test of lead wires (see dwg. 9206137) - Major defect - Code No. 03013.

4.3.3.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, eighty (80) assemblies shall be selected from each lot for test. The lot shall be rejected if two (2) or more assemblies fail to comply with the requirements. The test shall be performed in accordance with 4.4.3.

4.3.3.3.2 After three consecutive lots have complied with the criteria of 4.3.3.3.1, a single sampling plan in accordance with code letter G and AQL of 0.40 percent shall be used for each lot.

4.3.3.4 Dielectric strength, attenuation, and resistance tests (see Table III) -

TABLE III

Test	Code No.	Test Classification	No. of Samples	Test Paragraph
Dielectric strength (lead to lead) (see dwg. 9206137)	03015	Major	100%	4.4.4
R-F attenuation (see dwg. 9206137)	03016	Critical	100%	4.4.5
Lead circuit resistance (see dwg. 9206140)	07012	Major	100%	4.4.6
Resistance between shorted lead wires and cup (see dwg. 9206140)	07013	Major	100%	4.4.7
Resistance between lead wires (see dwg. 9206137)	03017	Major	100%	4.4.8

Any detonator assembly that fails to comply with a requirement shall be classed defective and removed from the lot.

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4.3.4 Inspection Equipment.- Equipment Tabulation Number ET - 9206140 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.- Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.- The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 Moisture content.

4.4.1.1 Lead azide.- The moisture content of the lead azide shall be determined in accordance with Method 101.5 of MIL-STD-650 except that a two gram (gr.) sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene.

4.4.1.2 Moisture content of PETN.- The moisture content of the PETN shall be determined as specified in 4.4.1.1 except that the drying temperature and time shall be 95 plus or minus 2 degrees C. for one hour.

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4.4.2 Functioning.- Detonator assemblies used for the following tests shall be temperature conditioned to 70 plus or minus 10 degrees Fahrenheit (F.) for 2 hours min. prior to testing.

4.4.2.1 Time.- The detonator assemblies shall be assembled in the test fixture specified in 4.3.4 and fired by applying the energy from a 16 microfarad capacitor charged to 2.5 volts max. The elapsed time between application of the energy and explosion shall be measured.

4.4.2.2 Waterproofness and Output.- The detonator shall be submerged in water for the time specified, and at the depth and temperature specified. After waterproofness, the detonator assemblies shall be assembled in the test fixture specified in 4.3.4 and fired by applying the same energy specified in 4.4.2.1. The diameter of the perforation made in the lead disc shall be measured.

4.4.2.3 Duds.- The detonator assemblies shall be assembled in the test fixture specified in 4.3.4 and fired by applying the same energy specified in 4.4.2.1. Observation shall be made to determine compliance with the requirements.

4.4.3 Pull test of lead wires.- The security of the lead wires in the plug shall be tested in accordance with the procedure and equipment specified in 4.3.4.

4.4.4 Dielectric strength lead to lead.- The dielectric strength between the leads shall be determined in accordance with the procedure and equipment specified in 4.3.4.

4.4.5 R-F attenuation test.- The R-F attenuation shall be determined in accordance with the procedure and equipment specified in 4.3.4.

4.4.6 Lead circuit resistance.- The resistance of the lead circuit shall be measured when a current limited to one milliampere is applied. The test equipment shall be in accordance with 4.3.4.

4.4.7 Resistance between shorted lead wires and cup.- The resistance shall be measured when 1.5 volts direct current limited to 50 microamperes is applied between lead wires and the cup. The test equipment shall be in accordance with 4.3.4.

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4.4.8 Resistance between lead wires.- The resistance between the lead wires shall be determined in accordance with the procedure and equipment specified in 4.3.4.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging.

5.1.1 Level B.

5.1.1.1 Rosette wrap.- Twenty-five (25) detonators with twisted lead wires out straight shall be made into a bundle using the lead wires of one detonator to secure the bundle. Each bundle shall be wrapped in approximately an eight (8) by eight (8) by .0015 inch thick size sheet of aluminum foil complying with Specification MIL-A-148, so that all detonators and lead wires in a bundle are completely encased in the foil. Five (5) bundles of detonators (125 detonators) shall be packed in a folded carton set-up box, or comparable container, made of material with a minimum board thickness of 0.03 inch. When destined for shipment and uncovered storage in excess of 180 days from the loading date shown on the packing box to the date of assemble into end items, the following shall be included: Each carton, set-up box, or comparable container, after blunting corners, shall be placed in a water and vapor proof bag manufactured in accordance with Type I, Class E, of Specification MIL-B-117. The bag shall be closed. All excess air shall be exhausted within the bag and the bag then sealed in accordance with the manufacturer's instructions. The seal shall be located at the edge of the bag to provide for a minimum of two subsequent seals, if required. The seal shall comply with the heat seal requirements of Specification MIL-P-116.

5.1.1.2 Pocket assembly, alternate pack.- The detonators shall be packed in pocket assemblies manufactured in accordance with Figure 1 (as applicable) made from .0015 inch thick aluminum foil complying with Specification MIL-A-148. An adhesive seal shall be used to bond the faces of the aluminum foil together to form five (5) pockets. The strength of the seal shall be sufficient to allow insertion of the dowel (see Figure 1) without separation of the seal. The size of each pocket between seals shall not vary by more than 1/16 inch. Each pocket shall be opened to form a cylinder

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using a smooth dowel with rounded end (see Figure 1). Any alternate method that results in a satisfactory pocket without damage to the material may be used. One M78 Electric Detonator assembly shall be inserted, detonator end down, in each pocket. The lip of the pocket assembly shall be folded over to close the pouch. Twenty-five (25) closed pocket assemblies containing a total of 125 detonators shall be packed in a folded carton, set-up box, or comparable container made of material with a minimum board thickness of 0.03 inch. If shipment and uncovered storage is to exceed 180 days, the detonator container will be protected as specified in 5.1.1.1.

5.1.2 Level C.- Unless otherwise specified, the M78 Electric Detonators shall be preserved and packaged as specified in 5.1.1 except that the barrier bag is not required.

5.2 Packing.

5.2.1 Level B.- A suitable number of containers each containing 125 detonator assemblies shall be packed in an outer case in accordance with Tariff 10, Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, paragraph 73.66 (see Code of Federal Regulations 49 CFR 71-78).

5.2.2 Level C.- Unless otherwise specified, M78 Electric Detonators shall be packed as specified in 5.2.1.

5.3 Marking.- Each carton shall be printed, stamped, or labeled and each barrier bag printed or stamped in accordance with the following information and instructions:

a. 125 Detonator, Electric, M78
Lot No. Date packed (Mo/Yr)

b. All letters shall be upper case. Size of letters and numerals shall be a min. of 1/4 inch high applied with black waterproof ink. Stencil ink shall conform to Specification TT-I-559.

c. Printing and labeling shall be commercial type. Labels shall be on white gummed paper.

d. Exterior boxes shall be marked in accordance with dwg. 8796522. The ICC designation shall be (see Note):
Electric Blasting Caps - Handle Carefully

e. Note: "The number of caps packed in one outer pack shall be shown in the marking."

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6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Submission of first article quantity.- Instructions as to the location of the facility at which the evaluation of the first article quantity shall be carried out, may be obtained from the contracting officer.

6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Equipment Lists.- Equipment part or identifying lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (SMUPA Form 1010) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 through 6.3.6. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3, 6.3.5 and 6.3.6 will be designated in the EL by the following codes. Other inspection equipment listed on the EL and not coded are defined in 6.3.4.

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MU - Army design, mandatory for use.
CD - Contractor design and/or commercial equipment on contractor furmat. (Requires submission for Government approval as specified in 6.3.7).

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Envelope drawing.- Envelope drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of envelope drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of envelope drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.- Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those

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portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.- Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

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6.4 Submission of test data and ammunition data cards.- In addition to the normal distribution of records, for loaded items procured by the Department of the Army, one copy of all functioning test data and ammunition data cards shall be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey
ATTN: SMUPA-ND5
SMUPA-DC

6.5 Calculation of standard deviation.- The following formula or equivalent shall be used in calculating the sample standard deviation:

$$S = \sqrt{\frac{\Sigma (X-\overline{X})^2}{n-1}}$$

where:

S = sample standard deviation
X = numerical value of individual observations
$\overline{X}$ = average value of the sample observations
n = sample size

6.6 Check test for possible deterioration.- The following test is designed to prevent detonators that have deteriorated being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the primers are assembled does not require that the check test be performed, the contracting officer should insert in the contract a provision for the performance of the check test. The contracting officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Two hundred (200) detonator assemblies shall be selected from each lot and tested as follows:

6.6.1 Functioning time and duds.

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6.6.1.1 Sampling.- Forty (40) detonators shall be subjected to the test specified in 4.4.2.1. Observation shall be made for the functioning time, and all duds that occur shall be recorded. Duds will be replaced by detonators from the sample of 200.

6.6.1.2 Rejection.- The lot shall be rejected if the mean functioning time ($\bar{X}$) plus 2.03 times the standard deviation (S) exceeds 120 microseconds. Duds shall not be used in the standard deviation calculations.

6.6.2 Waterproofness, Output and Duds.

6.6.2.1 Sampling.- Fifty (50) detonator assemblies shall be subjected to the test specified in 4.4.2.2. Observation shall be made of the output, and duds that occur shall be recorded. Duds shall be replaced by detonators from the sample of 200.

6.6.2.2 Rejection.- The lot shall be rejected if two or more detonator assemblies fail to comply with the disc perforation requirement.

6.6.3 Duds.

6.6.3.1 Sampling.- The remainder of the 200 detonator assemblies selected in 6.6 shall be subjected to the test specified in 4.4.2.3.

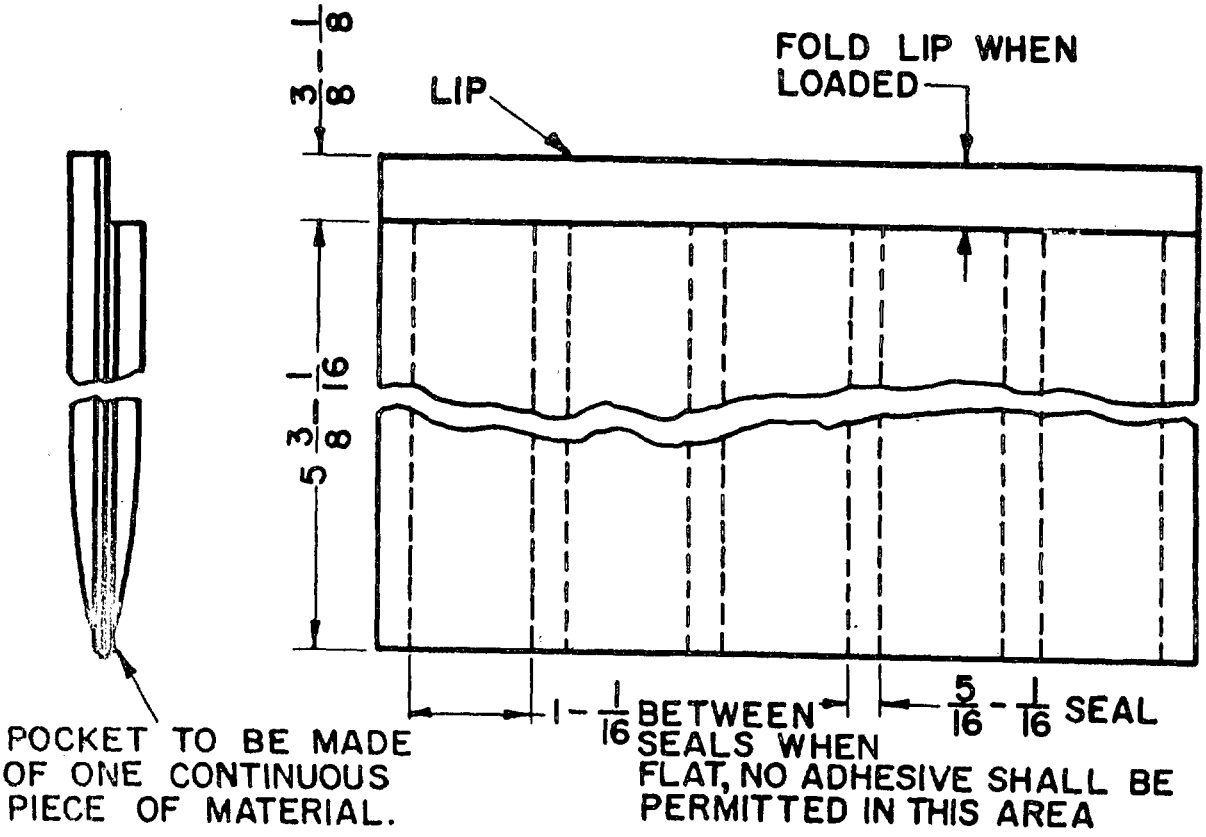
6.6.3.2 Rejection.- The lot shall be rejected if the number of detonator assemblies that fail to fire on the first attempt (duds) accumulated from 6.6.1.1, 6.6.2.1, and 6.6.3.1 equals or exceeds four (4).

Custodian:
Army-MU

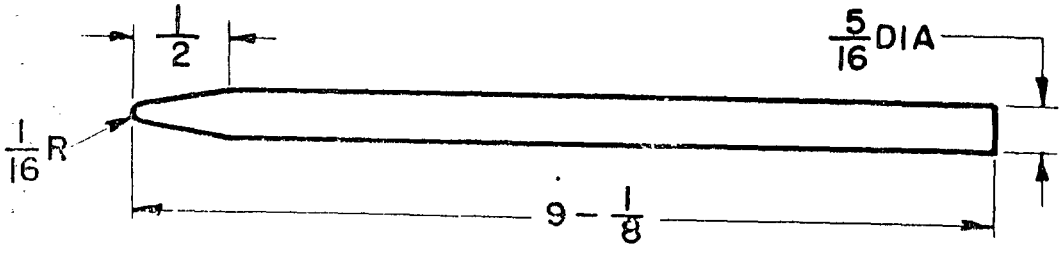
Preparing activity:
Army-MU

Project No. 1345-A-227

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POCKET ASSEMBLY



DOWEL SHALL BE SMOOTH WITH NO SHARP CORNERS

DOWEL

FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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